Robot field reports hold early warnings that nobody has time to read

inGen Field Intelligence · Chiyu Xu · Applied AI Analyst Intern

2,000

free-text reports from Aido Rover and Fari (synthetic, Jan–Jun 2026)

14

incident categories plus No Incident, with severity 1–4

4

hidden patterns planted and sealed, to test whether the analysis finds them

Four steps turn messy notes into tested, rate-based findings

1

Taxonomy

14 categories by where the robot failed, not the trigger; severity 1–4

2

Gold set

250 reports labelled by hand; 150 to tune, 100 held out and scored once

3

LLM classifier

Versioned prompt, JSON schema, retry then flag; cost logged per run

4

Analysis

Rates per operating hour, confidence intervals, multiple-testing correction

One command (python run_all.py) rebuilds every number and figure in this deck.

The LLM classifier is 96% accurate on unseen reports, ahead of both baselines

Category accuracy, 100 held-out reports



Human review: every confidence band met the 80% floor; only 1.6% of outputs are low-confidence.

Main error: a patrol paused by the weather rule is read as either Weather or No Incident (27 of 2,000). The fix is one sentence in the taxonomy, not a new model.

Against the generator's own labels: 96.9% on all 2,000 reports.

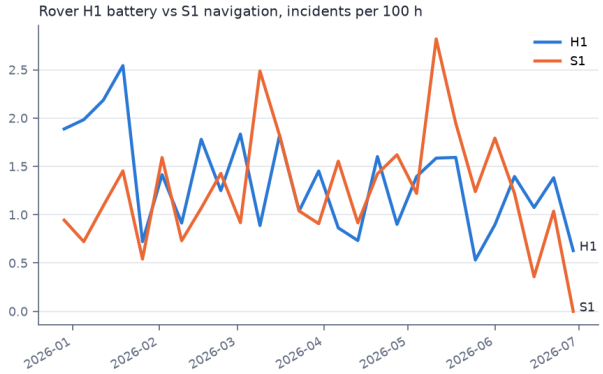
Cold nights and the April Fari update drive the biggest rises in incidents

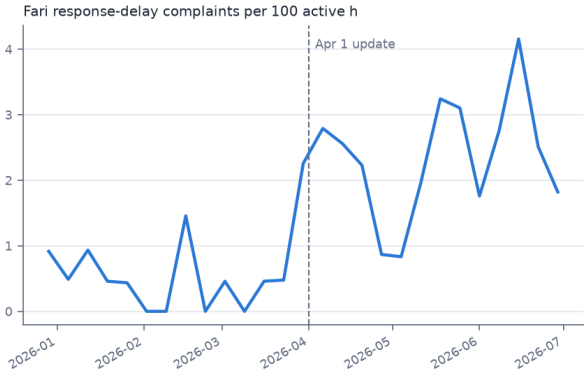
1.8×

battery incident rate below 0 °C (95% CI 1.32–2.45)

5.0×

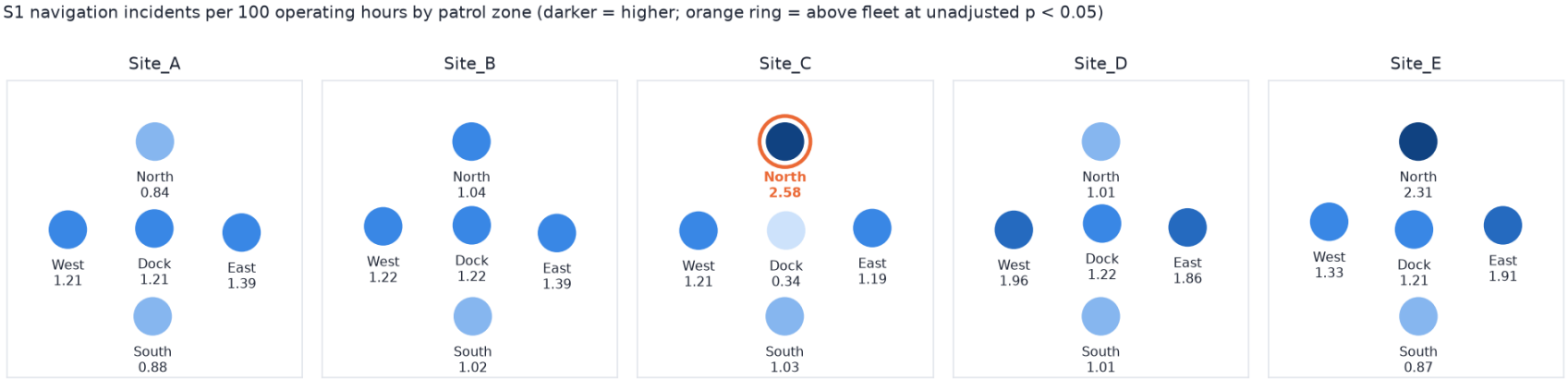
Fari response-delay complaints after 1 April (95% CI 2.76–9.07); latency 1.75 s → 3.04 s





Severity cannot be ranked yet: predicted severity matched hand labels only 47% of the time.

Site_C North has the most navigation incidents, but the data cannot yet confirm a hotspot



2.07×

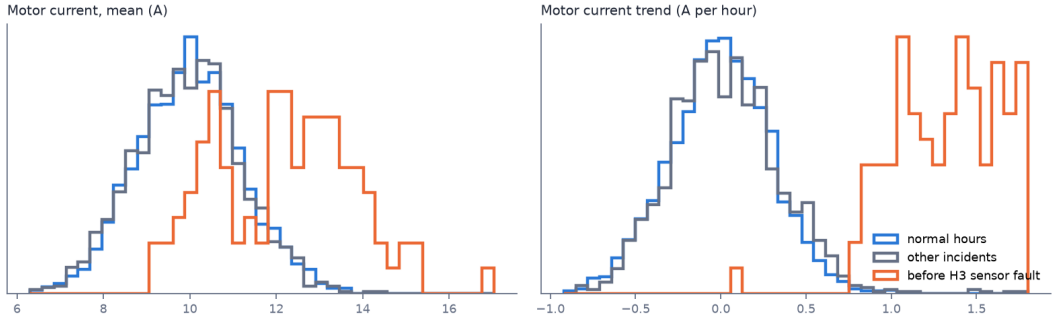
the fleet's navigation rate, highest of 25 zones

q = 0.33

after correcting for 25 zones: not significant

15 zone–category cells pass p < 0.05, about what 350 tests give by chance. Treat Site_C North as a lead to check, not a finding.

Rising motor current warns of sensor faults; most other incidents leave no trace



Signal in the hour before	AUC
Sensor fault: motor current trend	0.996
Battery: charge and drain	1.000
Navigation: localisation confidence	0.986
Obstacle, connectivity, app, access	≈ 0.5

0.996

cross-validated AUC: sensor-fault hours vs normal hours

Battery and navigation AUCs partly restate the label. The useful signal is the non-obvious one: the drive motor warns of a sensor fault.

These findings come from synthetic data, and one planted pattern was only partly found

Planted pattern	Result
Cold-weather battery incidents	Found
Fari delay complaints after update	Found
Motor current before sensor faults	Found
Site_C North navigation hotspot	Partial: sample too small

- **All data is synthetic**

Reports, operating hours and telemetry were generated, so results show what the method can detect.

- **One labeller**

Kappa 1.0 was a same-day re-label and overstates reliability.

- **The miss is sample size**

With the true labels the hotspot is still not significant.

Next: run the pipeline on real InGen reports and turn the sensor signal into an alert

1

Real reports

Run on a month of real field notes with names removed; label 50 per month to watch accuracy.

2

Sentinel Prime logs

Add alert logs as a second text source and link alerts to incidents.

3

Maintenance alert

Validate the motor-current signal on real telemetry, then alert on a rising trend.

Quick wins: fix the weather-pause rule; check the April Fari release for latency.